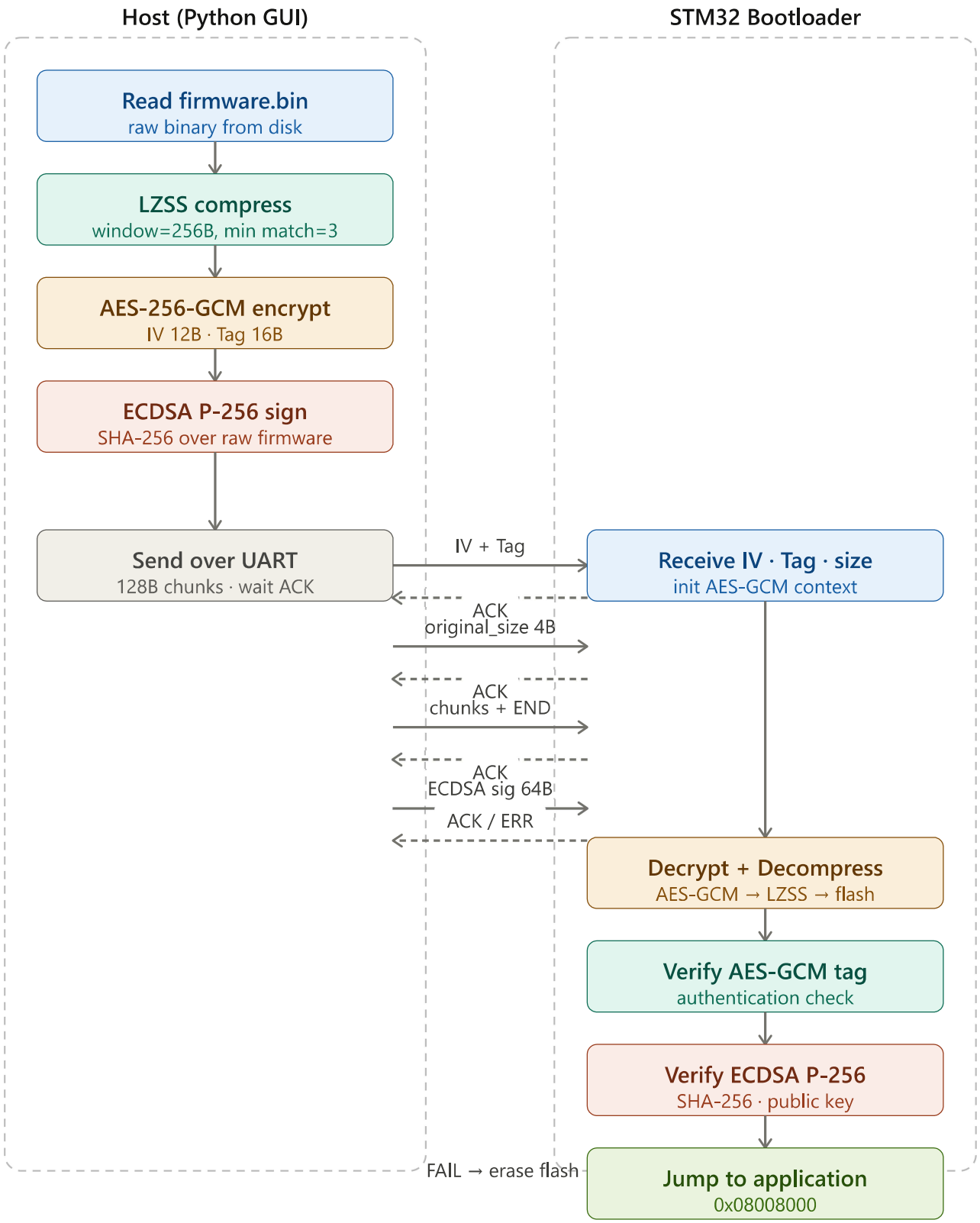
Host (Python GUI)
STM32 Bootloader
Read firmware.bin
raw binary from disk
LZSS compress
window=256B, min match=3
AES-256-GCM encrypt
IV 12B · Tag 16B
ECDSA P-256 sign
SHA-256 over raw firmware
Send over UART
128B chunks · wait ACK
IV + Tag
Receive IV · Tag · size
init AES-GCM context
ACK
original_size 4B
ACK
chunks + END
ACK
ECDSA sig 64B
ACK / ERR
Decrypt + Decompress
AES-GCM → LZSS → flash
Verify AES-GCM tag
authentication check
Verify ECDSA P-256
SHA-256 · public key
FAIL → erase flash
Jump to application
0x08008000